Creating a Consolidated/Container Database Manually

STEP 1: Create the database directory

- Create all the necessary directories (folders) for the database

- Create folders for the following files

- pfile (pfile),
- archive log files (recovery_area)
- control files (oradata)
- datafiles (oradata) –multiplex as needed
- redo log files (oradata) – multiplex as needed
- audit files (adump)
- pdbseed files (pdbseed)

Set path for diagnostic_dest parameter in file

folder for scripts

STEP 2: copy a pfile and create database script from an already built container database

Pfile should include

*.db_block_size=8192

*.open_cursors=300

*.db_name='cdb1'

*.control_files='C:\app\Administrator\oradata\cdb1\control01.ctl','C:\app\Administrator\recovery_area\cdb1\control02.ctl'

*.compatible='12.2.0'

*.diagnostic_dest='C:\app\Administrator'

*.enable_pluggable_database=true

*.nls_language='AMERICAN'

*.nls_territory='AMERICA'

*.local_listener='LISTENER_cdb1'

*.processes=320

*.sga_target=2000m

*.audit_file_dest='C:\app\Administrator\admin\cdb1\adump'

*.audit_trail='db'

*.remote_login_passwordfile='EXCLUSIVE'

*.dispatchers='(PROTOCOL=TCP) (SERVICE=cdb1XDB)'

*.pga_aggregate_target=500m

*.db_recovery_file_dest='C:\app\Administrator\recovery_area'

*.db_recovery_file_dest_size=14250m

STEP 3: Create password file (UNNECESSARY if operating system authentication is available)

Enter the following command in OS command prompt

```
C:\Windows\system32>ORADIM -NEW -SID CDB2 -STARTMODE MANUAL
Enter password for Oracle service user:
Instance created.
```

- Password argument is the password for SYS user

- Maximum number of entries (users accounts) to permit in file

- Oracle service user password was “Passw0rd”

STEP 4: Set ORACLE_SID in command prompt to instance SID

```
C:\Windows\system32>set oracle_sid=cdb2
```

STEP 5: Connect to the database instance

```
C:\Windows\system32>sqlplus /nolog

SQL*Plus: Release 12.2.0.1.0 Production on Fri Dec 10 02:22:51 2021

Copyright (c) 1982, 2016, Oracle.  All rights reserved.

SQL> conn sys as sysdba
Enter password:
Connected to an idle instance.
SQL>
```

- For OS system authentication no password is required so you can press the enter key
- For Password file authentication enter the SYS password declared in STEP 3

Alternative method for OS system authentication

```
C:\Windows\system32>sqlplus /nolog

SQL*Plus: Release 12.2.0.1.0 Production on Fri Dec 10 02:32:46 2021

Copyright (c) 1982, 2016, Oracle.  All rights reserved.

SQL> conn / as sysdba
Connected to an idle instance.
SQL>
```

STEP 6: Create spfile from pfile (NOTE: make sure all file paths mentioned in the pfile exist and are correct)

```
SQL> CREATE SPFILE FROM PFILE = 'C:\CDB2\pfile\initCDB2.ora';

File created.

SQL>
```

Perform this command every time the pfile is modified

STEP 7: Start database in nomount mode

```
SQL> STARTUP NOMOUNT
ORACLE instance started.

Total System Global Area 2097152000 bytes
Fixed Size                  8920832 bytes
Variable Size             620757248 bytes
Database Buffers         1459617792 bytes
Redo Buffers                7856128 bytes
SQL>
```

STEP 8: Run Database Create Script (NOTE: make sure all file paths mentioned in the create database script exist and are correct)

STEP 9: Run the following scripts in the container database CDB$ROOT

@?\rdbms\admin\catalog.sql
@?\rdbms\admin\catproc.sql
@?\rdbms\admin\utlrp.sql

Connect to CDB as SYSTEM user to run the following script in the CDB (check con_name)

@?/sqlplus/admin/pupbld.sql

- '?' is a shortcut for oracle_home in SQL*PLUS

STEP 10: Open the pdb seed, PDB$SEED, in read and write mode

Set parameter _oracle_script to so you can put PDB$SEED in read and write mode

NOTE: PDB$SEED must be closed and reopened in read and write mode

```
SQL> ALTER SESSION SET "_oracle_script"= TRUE;

Session altered.

SQL> show parameter "_oracle_script"

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
_oracle_script                       boolean     TRUE
SQL>
```

```
SQL> ALTER PLUGGABLE DATABASE PDB$SEED CLOSE IMMEDIATE;

Pluggable database altered.

SQL> ALTER PLUGGABLE DATABASE PDB$SEED OPEN READ WRITE;

Pluggable database altered.

SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ WRITE NO
SQL>
```

Alternatively, you can use the keyword FORCE to switch modes while the pdb is still open

```
SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
SQL>
SQL>
SQL> ALTER PLUGGABLE DATABASE PDB$SEED OPEN READ WRITE FORCE;

Pluggable database altered.

SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ WRITE NO
SQL>
```

STEP 11: Set container to PDB$ROOT and run the scripts below

```
SQL> SHOW CON_NAME

CON_NAME
------------------------------
CDB$ROOT
SQL>
SQL> ALTER SESSION SET CONTAINER = PDB$SEED;

Session altered.

SQL> SHOW CON_NAME

CON_NAME
------------------------------
PDB$SEED
SQL>
```

Run the following scripts

@?\rdbms\admin\catalog.sql
@?\rdbms\admin\catproc.sql
@?\rdbms\admin\utlrp.sql

Connect to CDB as SYSTEM user then set container to PDB$SEED and run the following script

@?/sqlplus/admin/pupbld.sql

```
SQL> conn system@cdb2/Or5cle#123
Connected.
SQL> alter session set container = pdb$seed;

Session altered.

SQL> show con_name

CON_NAME
------------------------------
PDB$SEED
SQL> show user
USER is "SYSTEM"
SQL> _
```

STEP 12: Verify components installed in the database on the root and pdbseed

```
SQL> COLUMN comp_name FORMAT A50
SQL> COLUMN status FORMAT A20
SQL>
SQL> SELECT comp_name, status FROM dba_registry;

COMP_NAME                                          STATUS
-------------------------------------------------- --------------------
Oracle Database Catalog Views                      VALID
Oracle Database Packages and Types                 VALID
Oracle XML Database                                VALID

3 rows selected.

SQL> SHOW CON_NAME

CON_NAME
------------------------------
PDB$SEED
SQL>
```

```
SQL> SELECT comp_name, status FROM dba_registry;

COMP_NAME                                          STATUS
-------------------------------------------------- --------------------
Oracle Database Catalog Views                      VALID
Oracle Database Packages and Types                 VALID
Oracle XML Database                                VALID

3 rows selected.

SQL> SHOW CON_NAME

CON_NAME
------------------------------
CDB$ROOT
SQL>
```

STEP 13: Shutdown database (make sure con_name is CDB$ROOT) and enable Oracle Service Automatic startup

```
SQL> SHOW CON_NAME

CON_NAME
------------------------------
CDB$ROOT
SQL>
SQL> SHUTDOWN IMMEDIATE
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
```

```
C:\Windows\system32>ORADIM -EDIT -SID CDB2 -STARTMODE AUTO -SRVCSTART SYSTEM -SPFILE

C:\Windows\system32>
```

STEP 14: Modify tsn.ora file in oracle_home admin to include new database listener to the database SID to enable logging into database using SQL*Plus

filepath = C:\app\Oracle\product\12.2.0\Oracle_Home1\network\admin to include new database listener to enable connections to the database from the client end (e.g. using SQL*Plus)

“LISTENER_CDB2” is a default service name initialized in the pfile to allow connection to the database service, CDB2, which allows connections to the database. “CDB2” is a manually created service name that can also allow connections to the database service and database. “CDB2” would be listed in the listener services list WHEN it is active, but LISTENER_CDB2 does not appear on the listener service list whether it is active or not

```
CDB2 =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = WIN-6HE0A9IDPK9.localdomain)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = CDB2)
    )
  )

LISTENER_CDB2 =
  (ADDRESS = (PROTOCOL = TCP)(HOST = WIN-6HE0A9IDPK9.localdomain)(PORT = 1521))
```

Creating a Pluggable database from a Seed PDB (PDB$SEED)

STEP 1: Connect to Database as SYS with SYSDBA Privileges

STEP 2: Check that you are connected to the root container CDB$ROOT and that PDB$SEED is opened in read only mode

```
SQL> SELECT name FROM v$database;

NAME
---------
CDB2

SQL> SHOW CON_NAME

CON_NAME
------------------------------
CDB$ROOT
SQL>
SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
SQL>
```

STEP 3: Make sure the appropriate directories (folders) exist. Then enter the CREATE PLUGGABLE DATABASE command for creating a pluggable database from a seed PDB. The admin user for the pluggable database is PDB_ADM. The newly created PDB1 is currently closed

```
SQL> CREATE PLUGGABLE DATABASE pdb1 ADMIN USER pdb_adm IDENTIFIED BY Password1
  2  DEFAULT TABLESPACE users
  3    DATAFILE 'C:\CDB2\oradata\pdb1\users01.dbf' SIZE 5M
  4    AUTOEXTEND ON NEXT 1M MAXSIZE 50M
  5  FILE_NAME_CONVERT=('C:\CDB2\oradata\pdbseed',
  6                     'C:\CDB2\oradata\pdb1');

Pluggable database created.
```

```
SQL> SELECT pdb_name, status
  2  FROM dba_pdbs
  3  ORDER BY pdb_name;

PDB_NAME             STATUS
-------------------- ----------
PDB$SEED             NORMAL
PDB1                 NEW

SQL>
SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 PDB1                           MOUNTED
SQL>
```

STEP 4: Open the PDB in read/write mode (use keyword FORCE if already opened)

```
SQL> ALTER PLUGGABLE DATABASE PDB1 OPEN READ WRITE;

Pluggable database altered.

SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 PDB1                           READ WRITE NO
SQL>
```

STEP 5: Verify the State of the PDBS

```
SQL> COLUMN pdb_name FORMAT A20
SQL>
SQL> SELECT pdb_name, status
  2  FROM dba_pdbs
  3  ORDER BY pdb_name;

PDB_NAME             STATUS
-------------------- ----------
PDB$SEED             NORMAL
PDB1                 NORMAL

SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 PDB1                           READ WRITE NO
SQL>
```

STEP 6 (OPTIONAL): If you want the PDB to automatically open on startup, you will need to save the state of the PDB after it has been opened in read/write mode

ALTER PLUGGABLE DATABASE pdb1 SAVE STATE;

```
SQL> SELECT COUNT(*) FROM DBA_PDB_SAVED_STATES;

  COUNT(*)
----------
         0

SQL> ALTER PLUGGABLE DATABASE pdb1 SAVE STATE;

Pluggable database altered.

SQL> SELECT COUNT(*) FROM DBA_PDB_SAVED_STATES;

  COUNT(*)
----------
         1
```

alter

STEP 7: Modify tnsnames.ora file to allow connections to pdb

```
PDB1 =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = WIN-6HE0A9IDPK9.localdomain)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = PDB1)
    )
  )
```

Cloning a Local PDB -- from the same Consolidated Database/(CDB)

STEP 1: Place the source pdb in read only mode

```
SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 PDB1                           READ WRITE NO
SQL>
SQL> ALTER PLUGGABLE DATABASE pdb1 OPEN READ ONLY FORCE;

Pluggable database altered.

SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 PDB1                           READ ONLY  NO
```

STEP 2: Make sure the necessary directories exist then Issue the Create PDB command

```
SQL> CREATE PLUGGABLE DATABASE clonepdb FROM pdb1
  2        FILE_NAME_CONVERT=('C:\CDB2\oradata\pdb1',
  3                           'C:\CDB2\oradata\clonepdb');

Pluggable database created.
```

STEP 3: Check status of PDBS

```
SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 PDB1                           READ ONLY  NO
         5 CLONEPDB                       MOUNTED
SQL>
SQL> SELECT pdb_name, status
  2  FROM dba_pdbs
  3  ORDER BY pdb_name;

PDB_NAME             STATUS
-------------------- ----------
CLONEPDB             NEW
PDB$SEED             NORMAL
PDB1                 NORMAL
```

STEP 4: Open the pluggable database in read/write mode and set the source pdb back to read/write mode

```
SQL> ALTER PLUGGABLE DATABASE clonepdb OPEN READ WRITE;

Pluggable database altered.

SQL> ALTER PLUGGABLE DATABASE pdb1 OPEN READ WRITE FORCE;

Pluggable database altered.
```

STEP 5: Show status of PDBS

```
SQL> SHOW PDBS

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 PDB1                           READ WRITE NO
         5 CLONEPDB                       READ WRITE NO
SQL>
SQL> SELECT pdb_name, status
  2  FROM dba_pdbs
  3  ORDER BY pdb_name;

PDB_NAME             STATUS
-------------------- ----------
CLONEPDB             NORMAL
PDB$SEED             NORMAL
PDB1                 NORMAL
```

STEP 6: Modify tnsnames.ora file to allow connections to PDB